## CHAD: ASSESSING EXTERNAL AND PUBLIC SECTOR DEBT SUSTAINABILITY

17. This assessment presents a joint IMF-World Bank debt sustainability analysis using the standard debt dynamics templates for low-income countries (LICs) to analyze, on the basis of available data, the sustainability of Chad's external and public sector debt.

### Debt sustainability assessment

18. Even though Chad's debt situation has improved appreciably in recent years, the outlook remains fragile and Chad faces a high risk of debt distress. While the baseline scenario appears sustainable, it entails significant risks, including the possibility of further delays in Chad's receipt of full debt relief under the Highly Indebted Poor Countries (HIPC) Initiative and the Multilateral Debt Relief Initiative. Moreover, the alternative scenarios and bound tests indicate that in the event of adverse shocks Chad's debt burden indicators would deteriorate substantially. In particular, the debt sustainability outlook would deteriorate significantly if export values were to drop through a sharper than expected decline in oil prices, or if fiscal discipline were not to prevail in the near to medium-term.

### Recent debt developments

19. Chad is eligible for the Highly Indebted Poor Countries (HIPC) Initiative. It reached the HIPC Decision Point in 2001, at which time creditors agreed to a common reduction factor of 30 percent, resulting in HIPC debt relief of $170 million in NPV terms. While the full relief will only be delivered at the HIPC Completion Point, Chad became eligible for interim HIPC relief, from the Paris Club creditors and the IMF, IDA, and the AfDB Group. But as the Completion Point is delayed, Chad's may reach the ceilings on the provision of interim HIPC relief for some creditors. Nonetheless, Chad's debt situation has improved considerable over the past few years, thanks to the strong growth in overall GDP stemming from the oil sector and the reduced reliance by the government on external financial support. At end-2005, its external public and publicly guaranteed (PPG) debt stood at CFA 880 billion ($1.6 billion) or 27 percent of GDP, which compares with CFA 732 billion but 53 percent of GDP at end-2002. In NPV terms, the debt was only two thirds this amount since it is largely concessional: 90 percent is due to multilateral institutions and the remainder to bilateral creditors (see Staff Report Table 8).

### Baseline assumptions

20. The economic baseline scenarios informing the DSAs are much influenced by the expected evolution of Chad's oil production. This is expected to peak in 2007, followed by a gradual decline to one fifth that level by 2030, based on current proven oil reserves. At the same time, oil prices are expected to decline. That, together with investment cost recovery by companies, results in a sharp decline in oil revenues for the government (see Box 1). Other key assumptions are shown in Tables 1a and 1b. The baseline assumes a steady increase in

non-oil GDP growth as oil production declines. The main source of growth is the agricultural sector which will benefit from the infrastructure investments financed from oil revenue. With the help of envisaged measures, non-oil revenues will grow as a percent of GDP, and compensate for the decline in oil revenues so that the non-oil primary balance can cover debt service. In sum, the baseline includes a substantial medium-term adjustment and subsequent continued consolidation. As for debt service, the baseline assumes that Chad will reach the HIPC Completion Point at end-2007.[30] The baseline scenario shows relatively large residuals in 2006 and 2007, which are the result of the assumption that much of the oil revenue will be used to accumulate reserves. In subsequent years, as these reserves are drawn down, the residual declines and turns negative.

Box 1. Chad: Macroeconomic Projections for the Baseline DSA 2006-2026

| | | | | | Averages | | |
|---|---|---|---|---|---|---|---|
| | 2006 | 2007 | 2008 | 2009 | 2006-2013 | 2014-2026 | 2006-2026 |
| Real GDP growth (%) | 1.3 | -1.2 | 7.4 | 0.8 | 2.5 | 4.5 | 3.7 |
| Non-oil real GDP growth (%) | 5.1 | 5.9 | 4.5 | 4.6 | 5.0 | 5.6 | 5.3 |
| Nominal GDP growth (%) | 10.2 | -10.5 | 3.9 | -2.2 | 2.7 | 6.9 | 5.3 |
| Nominal GDP level (billions of CFAF) | 3,420 | 3,060 | 3,178 | 3,108 | 3,357 | 5,953 | 4,964 |
| | | | | | | | |
| Oil production (millions of barrels) | 55.4 | 47.5 | 56.6 | 50.3 | 48.1 | 20.1 | 30.8 |
| Oil prices (Brent, $ per barrel) | 65.2 | 52.0 | 49.5 | 43.2 | 47.3 | 56.4 | 52.9 |
| | | | | | | | |
| Oil revenues (CFAF billions) | 400 | 619 | 151 | 219 | 292 | 219 | 247 |
| Oil revenues (% of GDP) | 11.7 | 20.2 | 4.8 | 7.0 | 8.8 | 4.2 | 5.9 |
| Non-oil revenues (CFAF billions) | 153 | 173 | 191 | 211 | 231 | 735 | 543 |
| Non-oil revenues (% of GDP) | 4.5 | 5.7 | 6.0 | 6.8 | 6.8 | 11.9 | 10.0 |
| Non-oil primary balance (% of non-oil GDP) (Excluding grants and security spending) | -12.3 | -13.4 | -12.1 | -10.8 | -9.1 | -3.6 | -5.7 |

Sources: Chadian authorities and Fund staff estimates.

## External public and publicly guaranteed debt DSA

21. For external public debt, the baseline scenario assumes that the large fiscal surpluses in 2007 and 2008 when oil revenue peaks are saved and drawn down in subsequent years in line with the economy's absorptive capacity. Throughout the forecast period, Chad continues to roll over maturing debt to help finance investment needs for poverty reduction identified in the PRSP, but borrows only on concessional terms. Moreover, Chad's anti-poverty policies would attract a steady flow of grants from external donors. The result is that the NPV of debt remains in a range of 15-20 percent of GDP, based on an average rate of growth of real GDP

[30] If Chad were to receive MDRI in 2008, the amount would be on the order $1.2 billion which translates into $574 million in NPV terms which is 9 percent of GDP and 18 percent of exports.

of about 4 percent for the period (see Tables 1a and 2a). Likewise, in terms of exports, the NPV value of debt behaves moderately, staying within a range of 50-100 percent.

7. The external debt sustainability analysis includes a comparison with two alternative scenarios, one where key variables are kept at their historical averages (calculated only over 2000-2005 due to data availability), and another where new public sector loans are contracted on less favorable terms. The NPV-of-debt ratios increase under these scenarios but not disconcertingly so (Table 1b). Under the first one, A1, the current account deficit continues at the high level of 35 percent level which it achieved in the past on average thanks to the build up of the oil sector. Even though direct investment inflows and real GDP growth are commensurately higher, a gradual build up of debt occurs. Scenario A2 assumes that the interest on new borrowing is 2 percentage points higher than in the baseline, and the result is a higher NPV value of the debt stock.

8. In addition to the alternative scenarios, the baseline projections for external public debt were subjected to six standard bound tests (Figure 1 and Table 1b). Not surprisingly, Chad's debt position is most sensitive to the external shock consisting of a one-standard deviation drop in exports proceeds in 2007 and 2008, as could result from a roughly 20 percent drop in oil prices (Test B2). The GDP drop—one standard deviation below the high historical average (Test B1)—does not show much change from the baseline, the reason being that real GDP growth in the baseline in 2007 and 2008 is already at that level.

**Public debt DSA**

9. In addition to its external public debt, Chad has only limited domestic public debt, equivalent to about 6 percent of GDP at end-2005, and the baseline expects the share of domestic debt in total to remain roughly the same (Table 2a). On that assumption, an analysis of the sustainability of total public debt confirms broadly the conclusions of the previous analysis, namely that Chad's debt situation remains fragile. If the assumptions of the baseline materialize, debt appears sustainable, with the NPV of debt remaining in the 20–30 percent range (Table 2b and Figure 2). The standard "no reform" alternative scenario would be "primary balance unchanged from 2006". However, given the very large and temporary surplus in that year, staff opted for a "no reform" scenario that shows primary expenditures remaining at their high 2007 level despite the subsequent decline in oil revenue (shown as "no reform" in Figure 2 and as A2 in Table 2b). The most extreme stress test turns out to be the scenario of "permanently lower GDP growth" (A3) which causes NPV of debt to GDP and to revenues to rise sharply.

**Threshold comparison**

10. A standard component of a LIC DSA is to assess the external debt burden indicators in relation to policy-dependent thresholds reflecting the key empirical finding that a LIC with better policies and institutions can sustain a higher level of external debt. The LIC DSA framework therefore classifies countries into three performance categories—strong, medium, poor—as assessed by the World Bank's Country Policy and Institutional Assessment (CPIA).

Corresponding to these categories, the framework establishes three indicative thresholds for each debt burden indicator. Chad's CPIA score in 2005 was 2.9 which is in the bottom quintile and therefore characterizes Chad as a poor reformer with correspondingly low threshold values (Box 2).[31] As noted above, under the baseline scenarios most indicators remain below these thresholds. However, several alternative scenarios and bound tests produce ratios that breach those thresholds (see Figures 1 and 2). In addition serious risks attach to the baseline scenario, especially whether Chad will proceed as rapidly to the HIPC Completion Point as envisaged, and whether fiscal discipline will prevail in the near term. On balance, therefore, Chad appears to face a high risk of debt distress.

Box 2. Chad: Summary of Baseline Debt Sustainability Indicators 1/

| | Indicative Threshold 2/ | 2006 | 2016 | 2026 | Average 2006-26 |
|---|---|---|---|---|---|
| NPV of debt to GDP | 30 | 16 | 19 | 15 | 18 |
| NPV of debt to exports | 100 | 27 | 75 | 113 | 75 |
| NPV of debt to revenue | 200 | 59 | 68 | 69 | 75 |
| Debt service to exports | 15 | 1 | 3 | 7 | 4 |
| Debt service to revenue | 25 | 4 | 3 | 5 | 5 |

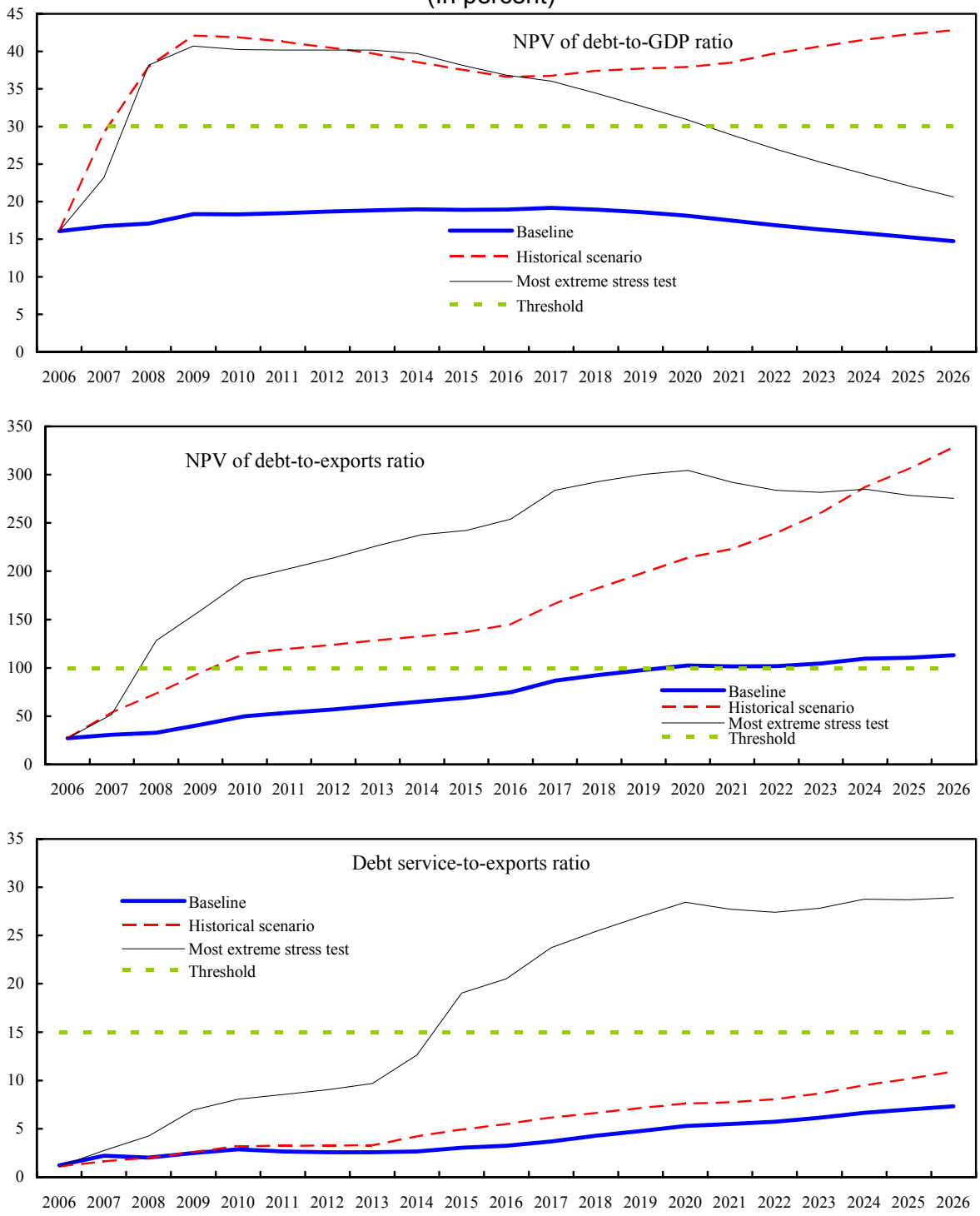
1./ The thresholds used for Chad indicate the level of debt indicators at which countries with a similar evaluation of policies and institutions would have at least a 25 percent chance of having a prolonged incident of debt distress in the coming year. Chad lies with the bottom quintile of countries ranked by the World Bank's Country Policy and Institutional Assessment Index (CPIA).

## Conclusions

11. Chad's debt situation has improved in recent years but remains fragile and the country still faces a high risk of debt distress. To further solidify the debt situation, it is of utmost importance that Chad reaches soon the HIPC Completion Point. This will require maintaining macroeconomic stability in the face of sharply fluctuating oil revenues in the near to medium term. It also requires speedy progress with the structural reform agenda reflected in the HIPC Completion Point Triggers. Such policies would permit the gradual diversification of the economy that is essential to ensure growth over the long term.

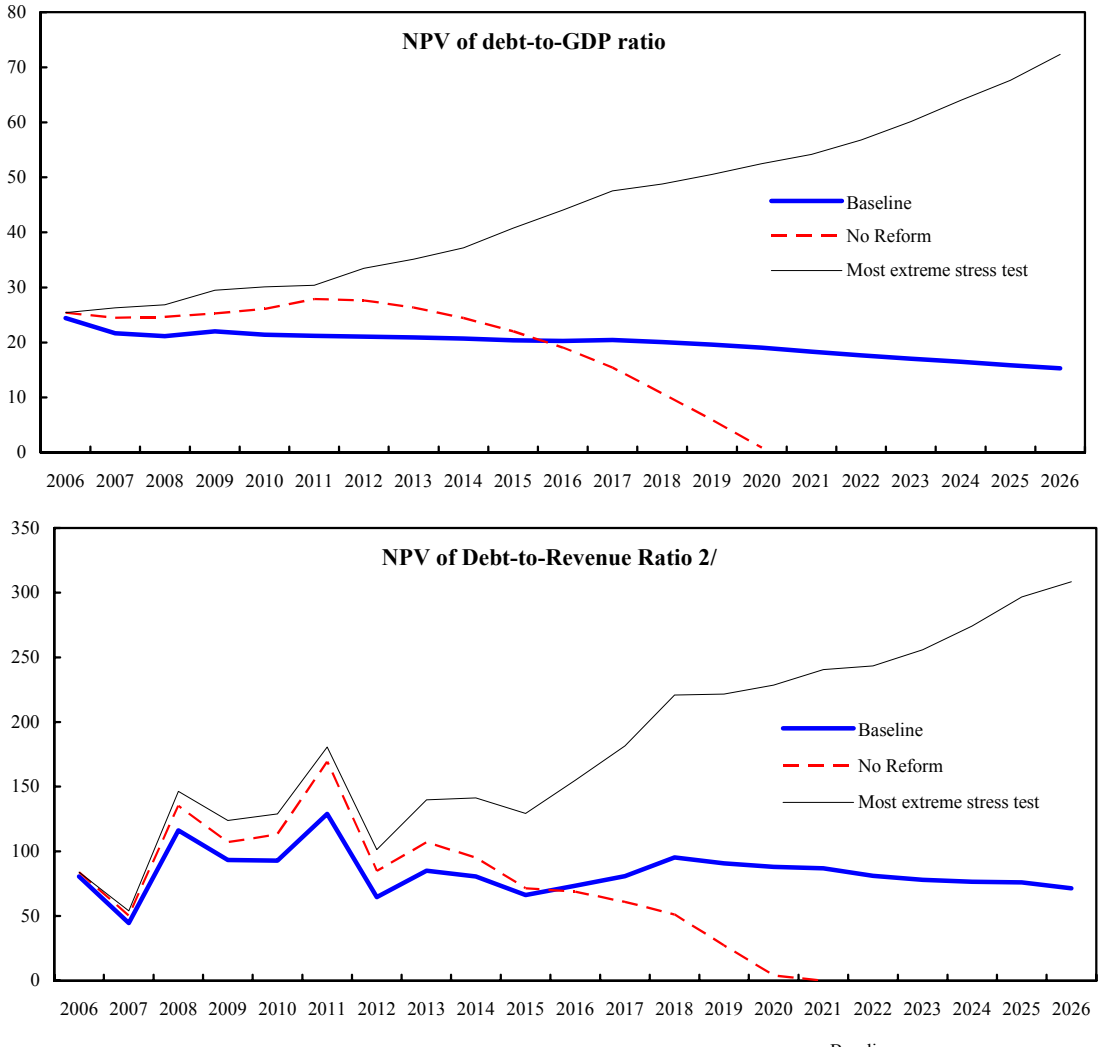
[31] Using the 2003-05 average of the CPIA would not change Chad's performance category in terms of policies and institutions. Information on the CPIA 2005 is now publicly available on the World Bank website at http://web.worldbank.org/WBSITE/EXTERNAL/EXTABOUTUS/IDA/0,,contentMDK:20933600~menuPK:2626968~pagePK:51236175~piPK:437394~theSitePK:73154,00.html

Figure 1. Chad: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2006-26
(In percent)




Source: Staff projections and simulations.
1./ Most extreme stress test is test that yields highest ratio in 2016 (B2 in all three cases).

Figure 2.Chad: Indicators of Public Debt Under Alternative Scenarios, 2006-2026 1/




Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2016 (A3, A3 and B3 respectively).
2/ Revenue including grants.

Table 1a. Chad: External Debt Sustainability Framework, Baseline Scenario, 2006-2026 1/
(In percent of GDP, unless otherwise indicated)

| | | Actual | | | Historical | Standard | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | # | 2003 | 2004 | 2005 | Average 6/ | Deviation 6/ | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | 2006-11 Average | 2016 | 2026 | 2012-26 Average |
| **External debt (nominal) 1/** | **...** | **50.2** | **35.0** | **27.0** | | | **20.5** | **24.1** | **24.4** | **26.0** | **25.8** | **26.0** | | **27.6** | **27.2** | |
| o/w public and publicly guaranteed (PPG) | ... | 50.2 | 35.0 | 27.0 | | | 20.5 | 24.1 | 24.4 | 26.0 | 25.8 | 26.0 | | 27.6 | 27.2 | |
| Change in external debt | ... | -6.8 | -15.2 | -8.0 | | | -6.5 | 3.6 | 0.3 | 1.6 | -0.3 | 0.2 | | 0.5 | -0.3 | |
| Identified net debt-creating flows | ... | 8.6 | -22.2 | -18.2 | | | -10.4 | -10.7 | -7.6 | -1.6 | 3.6 | 4.3 | | 5.7 | 1.6 | |
| **Non-interest current account deficit** | **...** | **49.5** | **7.5** | **0.7** | **35.2** | **33.5** | **-0.9** | **-3.5** | **2.7** | **6.3** | **11.2** | **11.3** | 4.5 | **9.7** | **3.5** | 6.8 |
| Deficit in balance of goods and services | ... | -9.3 | -35.8 | -45.9 | | | -50.1 | -44.6 | -41.8 | -33.6 | -25.8 | -23.6 | | -14.4 | -3.4 | |
| Exports | ... | 24.8 | 51.0 | 54.9 | | | 58.9 | 54.4 | 51.8 | 44.5 | 36.6 | 34.5 | | 25.3 | 13.0 | |
| Imports | ... | 15.4 | 15.2 | 9.1 | | | 8.8 | 9.9 | 10.0 | 10.8 | 10.8 | 10.9 | | 10.9 | 9.6 | |
| Net current transfers (negative = inflow) | ... | -3.6 | -4.9 | -3.1 | -3.6 | 2.0 | -2.1 | -3.1 | -3.0 | -3.0 | -2.8 | -2.7 | -2.8 | -2.0 | -4.0 | -3.4 |
| Other current account flows (negative = net inflow) | ... | 62.5 | 48.2 | 49.6 | | | 51.3 | 44.1 | 47.4 | 42.9 | 39.8 | 37.6 | | 26.0 | 10.9 | |
| **Net FDI (negative = inflow)** | **...** | **-26.2** | **-10.8** | **-10.4** | **-21.3** | **14.6** | **-9.5** | **-7.8** | **-8.9** | **-8.1** | **-7.3** | **-6.6** | -8.0 | **-3.5** | **-0.7** | -2.5 |
| **Endogenous debt dynamics 2/** | **...** | **-14.7** | **-18.9** | **-8.5** | | | **0.0** | **0.6** | **-1.4** | **0.2** | **-0.3** | **-0.4** | | **-0.5** | **-1.1** | |
| Contribution from nominal interest rate | ... | 0.6 | 0.4 | 0.2 | | | 0.3 | 0.3 | 0.3 | 0.4 | 0.3 | 0.3 | | 0.4 | 0.3 | |
| Contribution from real GDP growth | ... | -6.1 | -10.4 | -2.3 | | | -0.3 | 0.3 | -1.7 | -0.2 | -0.7 | -0.7 | | -0.8 | -1.4 | |
| Contribution from price and exchange rate changes | ... | -9.2 | -8.8 | -6.5 | | | … | … | … | … | … | … | | … | … | |
| **Residual (3-4) 3/** | **...** | **-15.4** | **6.9** | **10.3** | | | **3.9** | **14.3** | **7.9** | **3.2** | **-3.8** | **-4.1** | | **-5.2** | **-2.0** | |
| o/w exceptional financing | ... | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| NPV of external debt 4/ | ... | ... | ... | 17.9 | | | 16.1 | 16.7 | 17.1 | 18.4 | 18.3 | 18.5 | | 18.9 | 14.8 | |
| In percent of exports | ... | ... | ... | 32.5 | | | 27.3 | 30.7 | 33.0 | 41.2 | 50.1 | 53.5 | | 74.9 | 113.2 | |
| **NPV of PPG external debt** | **...** | **...** | **...** | **17.9** | | | **16.1** | **16.7** | **17.1** | **18.4** | **18.3** | **18.5** | | **18.9** | **14.8** | |
| In percent of exports | ... | ... | ... | 32.5 | | | 27.3 | 30.7 | 33.0 | 41.2 | 50.1 | 53.5 | | 74.9 | 113.2 | |
| Debt service-to-exports ratio (in percent) | ... | 5.8 | 1.7 | 1.5 | | | 1.2 | 2.2 | 2.0 | 2.5 | 2.9 | 2.6 | | 3.2 | 7.3 | |
| **PPG debt service-to-exports ratio (in percent)** | ... | 5.8 | 1.7 | 1.5 | | | 1.2 | 2.2 | 2.0 | 2.5 | 2.9 | 2.6 | | 3.2 | 7.3 | |
| Total gross financing need (billions of U.S. dollars) | ... | 0.7 | -0.1 | -0.5 | | | -0.6 | -0.6 | -0.3 | 0.0 | 0.3 | 0.4 | | 0.6 | 1.6 | |
| Non-interest current account deficit that stabilizes debt ratio | ... | 56.3 | 22.7 | 8.7 | | | 5.6 | -7.1 | 2.3 | 4.7 | 11.5 | 11.1 | | 9.2 | 3.8 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | ... | 14.7 | 33.6 | 8.6 | **12.7** | **11.5** | 1.3 | -1.2 | 7.4 | 0.8 | 2.7 | 2.7 | 2.3 | 3.2 | 5.7 | 4.3 |
| GDP deflator in US dollar terms (change in percent) | ... | 19.2 | 21.3 | 22.8 | **12.0** | **11.9** | 9.1 | -7.4 | -2.8 | -2.6 | 3.4 | 2.3 | 0.3 | 2.0 | 2.9 | 2.3 |
| Effective interest rate (percent) 5/ | ... | 1.4 | 1.2 | 0.8 | **1.3** | **0.3** | 1.2 | 1.5 | 1.5 | 1.5 | 1.3 | 1.3 | 1.4 | 1.4 | 1.2 | 1.3 |
| Growth of exports of G&S (US dollar terms, in percent) | ... | 166.9 | 233.9 | 43.6 | **72.6** | **103.1** | 18.5 | -15.4 | -0.7 | -15.7 | -12.7 | -0.8 | -4.5 | -2.9 | 2.8 | 0.0 |
| Growth of imports of G&S (US dollar terms, in percent) | ... | 48.1 | 59.4 | -20.3 | **24.0** | **33.7** | 7.0 | 3.0 | 6.1 | 6.1 | 5.4 | 6.0 | 5.6 | 5.5 | 6.5 | 5.8 |
| Grant element of new public sector borrowing (in percent) | | ... | ... | ... | ... | ... | 34.6 | 35.8 | 35.8 | 35.8 | 35.8 | 35.8 | 35.6 | 35.8 | 35.8 | 35.8 |
| *Memorandym item:* | | | | | | | | | | | | | | | | |
| Nominal GDP (billions of US dollars) | ... | 2.7 | 4.4 | 5.9 | | | 6.5 | 6.0 | 6.2 | 6.1 | 6.5 | 6.8 | | 8.8 | 18.0 | |

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and $\rho$ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are derived over the period 2000-2005 due to data availability.

Table 1b. Chad: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006-26
(In percent)

| | | | Projections | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | **2016** | 2026 |
| **NPV of debt-to-GDP ratio** | | | | | | | | |
| **Baseline** | 16 | 17 | 17 | 18 | 18 | 18 | **19** | 15 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2007-26 1/ | 16 | 29 | 38 | 42 | 42 | 41 | **37** | 43 |
| A2. New public sector loans on less favorable terms in 2007-26 2/ | 16 | 17 | 18 | 20 | 20 | 21 | **24** | -6 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2007-08 | 16 | 16 | 18 | 19 | 19 | 19 | **20** | 15 |
| B2. Export value growth at historical average minus one standard deviation in 2007-08 3/ | 16 | 23 | 38 | 41 | 40 | 40 | **37** | 21 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08 | 16 | 15 | 15 | 16 | 16 | 17 | **17** | 13 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/ | 16 | 18 | 20 | 21 | 21 | 21 | **21** | 15 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 16 | 0 | -19 | -20 | -19 | -19 | **-13** | 2 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/ | 16 | 25 | 25 | 27 | 27 | 27 | **28** | 22 |
| **NPV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 27 | 31 | 33 | 41 | 50 | 54 | **75** | 113 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2007-26 1/ | 27 | 54 | 73 | 95 | 115 | 120 | **145** | 328 |
| A2. New public sector loans on less favorable terms in 2007-26 2/ | 27 | 32 | 35 | 45 | 56 | 61 | **95** | -49 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2007-08 | 27 | 31 | 33 | 41 | 50 | 54 | **75** | 113 |
| B2. Export value growth at historical average minus one standard deviation in 2007-08 3/ | 27 | 52 | 128 | 159 | 191 | 203 | **254** | 275 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08 | 27 | 31 | 33 | 41 | 50 | 54 | **75** | 113 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/ | 27 | 32 | 38 | 47 | 57 | 61 | **83** | 118 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 27 | 1 | -28 | -35 | -41 | -42 | **-39** | 13 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/ | 27 | 31 | 33 | 41 | 50 | 54 | **75** | 113 |
| **Ratio of debt service to exports** | | | | | | | | |
| **Baseline** | 1 | 2 | 2 | 2 | 3 | 3 | **3** | 7 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2007-26 1/ | 1 | 2 | 2 | 3 | 3 | 3 | **5** | 11 |
| A2. New public sector loans on less favorable terms in 2007-26 2/ | 1 | 2 | 2 | 3 | 3 | 3 | **6** | 4 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2007-08 | 1 | 2 | 2 | 3 | 3 | 3 | **6** | 10 |
| B2. Export value growth at historical average minus one standard deviation in 2007-08 3/ | 1 | 3 | 4 | 7 | 8 | 9 | **21** | 29 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08 | 1 | 2 | 2 | 3 | 3 | 3 | **6** | 10 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/ | 1 | 2 | 2 | 3 | 3 | 3 | **6** | 11 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 1 | 2 | 1 | 0 | 0 | 0 | **-4** | -1 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/ | 1 | 2 | 2 | 3 | 3 | 3 | **6** | 10 |
| *Memorandum item:* | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 33 | 33 | 33 | 33 | 33 | 33 | **33** | 33 |

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a.Chad: Public Sector Debt Sustainability Framework, Baseline Scenario, 2003-2026

(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average [1/] | Standard Deviation [1/] | Estimate | Projections | | | | | Average 2006-11 | | | Average 2012-26 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2003 | 2004 | 2005 | | | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | | 2016 | 2026 | |
| **Public sector debt** [2/] | 48.2 | 41.9 | 36.0 | | | 26.6 | 29.1 | 28.5 | 29.7 | 28.9 | 28.7 | | 29.0 | 27.7 | |
| o/w foreign-currency denominated | 46.2 | 32.4 | 28.4 | | | 20.2 | 24.1 | 24.4 | 26.0 | 25.7 | 25.9 | | 27.5 | 27.1 | |
| Change in public sector debt | -6.3 | -6.2 | -6.0 | | | -9.4 | 2.4 | -0.5 | 1.2 | -0.8 | -0.1 | | 0.3 | -0.4 | |
| Identified debt-creating flows | -9.6 | -18.5 | -12.0 | | | -17.4 | -25.8 | -0.5 | -3.5 | -7.8 | -1.8 | | -9.2 | -2.2 | |
| Primary deficit | 5.3 | -0.5 | -4.9 | 2.4 | 4.2 | -12.7 | -28.9 | 0.7 | -4.1 | -6.2 | -0.6 | -8.6 | -8.1 | -0.3 | -3.7 |
| Revenue and grants | 16.1 | 14.5 | 17.6 | | | 30.3 | 48.7 | 18.2 | 23.6 | 23.0 | 16.5 | | 27.6 | 21.4 | |
| *of which* : grants | 8.3 | 3.2 | 3.6 | | | 2.5 | 2.6 | 2.7 | 2.7 | 2.7 | 2.8 | | 2.8 | 3.1 | |
| Primary (noninterest) expenditure | 21.4 | 14.0 | 12.7 | | | 17.6 | 19.9 | 18.9 | 19.5 | 16.9 | 15.9 | | 19.6 | 21.1 | |
| Automatic debt dynamics | -14.2 | -17.7 | -6.9 | | | -4.7 | 3.5 | -0.7 | 1.0 | -1.3 | -1.0 | | -1.0 | -2.0 | |
| Contribution from interest rate/growth differential | -7.3 | -12.6 | -4.9 | | | -1.0 | 1.2 | -1.8 | -0.2 | -1.0 | -0.9 | | -1.0 | -1.2 | |
| *of which* : contribution from average real interest rate | -0.4 | -0.5 | -1.6 | | | -0.6 | 0.9 | 0.2 | 0.0 | -0.2 | -0.1 | | -0.1 | 0.3 | |
| *of which* : contribution from real GDP growth | -7.0 | -12.1 | -3.3 | | | -0.5 | 0.3 | -2.0 | -0.2 | -0.8 | -0.8 | | -0.9 | -1.5 | |
| Contribution from real exchange rate depreciation | -6.9 | -5.1 | -2.0 | | | -3.7 | 2.3 | 1.0 | 1.1 | -0.3 | -0.1 | | ... | ... | |
| Other identified debt-creating flows | -0.6 | -0.3 | -0.3 | | | 0.0 | -0.4 | -0.4 | -0.3 | -0.3 | -0.3 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | -0.6 | -0.3 | -0.3 | | | 0.0 | -0.4 | -0.4 | -0.3 | -0.3 | -0.3 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | 3.3 | 12.2 | 6.0 | | | 8.1 | 28.3 | -0.1 | 4.6 | 7.0 | 1.7 | | 9.5 | 1.8 | |
| **NPV of public sector debt** | 22.6 | 30.9 | 29.0 | | | 24.5 | 21.7 | 21.2 | 22.0 | 21.4 | 21.2 | | 20.3 | 15.3 | |
| o/w foreign-currency denominated | 20.6 | 21.4 | 21.4 | | | 18.0 | 16.7 | 17.0 | 18.3 | 18.2 | 18.4 | | 18.9 | 14.7 | |
| o/w external | 20.6 | 21.4 | 21.4 | | | 18.0 | 16.7 | 17.0 | 18.3 | 18.2 | 18.4 | | 18.9 | 14.7 | |
| NPV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need [3/] | 15.3 | 6.6 | 4.7 | | | 2.7 | -4.0 | 10.0 | 7.3 | 4.6 | 6.5 | | 3.1 | 5.8 | |
| NPV of public sector debt-to-revenue ratio (in percent) [4/] | 139.8 | 212.6 | 165.0 | | | 80.6 | 44.5 | 116.1 | 93.3 | 92.9 | 128.8 | | 73.3 | 71.6 | |
| o/w external | 127.7 | 147.3 | 121.8 | | | 59.4 | 34.3 | 93.6 | 77.6 | 79.2 | 111.9 | | 68.2 | 68.7 | |
| Debt service-to-revenue ratio (in percent) [4/ 5/] | 10.8 | 7.3 | 7.6 | | | 4.2 | 4.2 | 10.2 | 7.1 | 6.4 | 7.7 | | 3.3 | 4.6 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | 11.6 | 5.7 | 1.1 | | | -3.4 | -31.3 | 1.2 | -5.3 | -5.3 | -0.4 | | -8.4 | 0.1 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 14.7 | 33.6 | 8.6 | 12.7 | 11.5 | 1.3 | -1.2 | 7.4 | 0.8 | 2.7 | 2.7 | 2.3 | 3.2 | 5.7 | 4.3 |
| Average nominal interest rate on forex debt (in percent) | 1.2 | 1.1 | 1.0 | 1.4 | 0.4 | 1.1 | 1.5 | 1.5 | 1.5 | 1.3 | 1.3 | 1.4 | 1.4 | 1.2 | 1.3 |
| Average real interest rate on domestic currency debt (in percent) | 4.8 | -3.6 | -17.2 | 0.8 | 0.8 | -6.1 | 12.8 | 5.9 | 4.6 | -0.2 | 1.9 | 3.2 | 1.5 | 0.5 | 1.2 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | -15.0 | -14.9 | -6.7 | -7.6 | 10.1 | -13.2 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | -0.4 | 10.3 | 22.6 | 8.9 | 8.5 | 8.8 | -9.4 | -3.3 | -3.0 | 2.9 | 1.5 | -0.4 | 2.0 | 2.9 | 2.3 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 26.7 | -12.5 | -1.4 | 6.9 | 15.9 | 40.4 | 11.3 | 2.1 | 4.2 | -11.1 | -3.4 | 7.3 | 5.6 | 6.2 | 6.3 |
| Grant element of new external borrowing (in percent) | 0.6 | 0.6 | 0.6 | 0.6 | 0.0 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 | 0.6 |

Sources: Country authorities; and Fund staff estimates and projections.

1/ Historical averages and standard deviations derived over the period 2000-2005 due to data availability.

2/Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.

3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

4/ Revenues including grants.

5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Table 2b. Chad: Sensitivity Analysis for Key Indicators of Public Debt 2006-2026

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2006 | 2007 | 2008 | 2009 | 2010 | 2011 | **2016** | 2026 |
| | .NPV of Debt-to-GDP Ratio | | | | | | | |
| **Baseline** | 24 | 22 | 21 | 22 | 21 | 21 | **20** | 15 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 25 | 47 | 45 | 47 | 47 | 44 | **49** | 27 |
| A2. No change in primary expenditures from 2007 | 25 | 24 | 25 | 25 | 26 | 28 | **19** | -25 |
| A3. Permanently lower GDP growth 1/ | 25 | 26 | 27 | 29 | 30 | 30 | **44** | 72 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008 | 25 | 23 | 24 | 25 | 24 | 22 | **12** | -16 |
| B2. Primary balance is at historical average minus one standard deviations in 2007-2008 | 25 | 29 | 34 | 36 | 35 | 34 | **25** | -7 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 25 | 52 | 54 | 56 | 53 | 51 | **36** | 0 |
| B4. One-time 30 percent real depreciation in 2007 | 25 | 27 | 26 | 26 | 24 | 21 | **6** | -29 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2007 | 25 | 34 | 33 | 34 | 31 | 30 | **16** | -18 |
| | NPV of Debt-to-Revenue Ratio 2/ | | | | | | | |
| **Baseline** | 81 | 44 | 116 | 93 | 93 | 129 | **73** | 72 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 84 | 98 | 256 | 206 | 213 | 287 | **188** | 145 |
| A2. No change in primary expenditures from 2007 | 84 | 50 | 135 | 107 | 113 | 169 | **69** | -115 |
| A3. Permanently lower GDP growth 1/ | 84 | 54 | 146 | 124 | 129 | 181 | **155** | 309 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2007-2008 | 84 | 47 | 131 | 105 | 102 | 134 | **45** | -75 |
| B2. Primary balance is at historical average minus one standard deviations in 2007-2008 | 84 | 60 | 188 | 152 | 151 | 205 | **90** | -32 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 84 | 106 | 299 | 239 | 233 | 317 | **131** | 0 |
| B4. One-time 30 percent real depreciation in 2007 | 84 | 56 | 142 | 110 | 103 | 129 | **21** | -136 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2007 | 84 | 69 | 180 | 142 | 137 | 180 | **58** | -84 |

Sources: Country authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).

2/ Revenues are defined inclusive of grants.